Ashutosh Ghimire

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EDUCATION

PhD in Computer Science and Engineering **Present**
Wright State University, Dayton, Ohio, USA (GPA: 3.8 out of 4)
Master of Science in Computer Science **April 2024**
Wright State University, Dayton, Ohio, USA (GPA: 3.5 out of 4
Bachelor of Engineering in Computer Engineering **February 2019**
Tribhuvan University, Kathmandu, Nepal

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, *Wright State University* **Dayton, OH | May 2025 – Present**

- Designed, trained, and evaluated machine learning pipelines on large-scale side-channel datasets, improving robustness by 77% under adversarial conditions, with deployment-ready inference workflows.
- Built an end-to-end ML pipeline (feature extraction > clustering > adversarial evaluation) in Python (scikit-learn) to systematically compare unsupervised models across attack strengths, eliminating inconsistent evaluation results observed in earlier experiments.
- Refactored data preprocessing and inference stages to parallelize feature fusion and batch clustering, reducing full experimental runtime while preserving deterministic outputs required for reproducible results.

Research Associate, *Wright State University* **Dayton, OH | August 2024 – April 2025**

- Developed adaptive ensemble learning models for real-time financial fraud detection after single-model approaches failed under concept drift, improving detection stability across evolving patterns.
- Integrated explainability techniques to interpret model decisions, improving analyst trust and usability in security-critical environments.
- Authored multiple whitepapers detailing research findings and advancements, with one notable paper endorsed by the National Institutes of Health (NIH) funding with a $100,000 grant provided to SMART Lab.
- Managed daily operations and project coordination for a lab of 13 students, training four new students during their onboarding phase and leading regular meetings, using Microsoft Office tools for documentation and project tracking.

Graduate Research Assistant, *Wright State University* **Dayton, OH | September 2022 – April 2024**

- Implemented a distributed federated learning system for secure AI hardware after centralized training exposed sensitive device-level data, achieving up to 3x speedup over baseline federated implementations.
- Devised an improved parallel approach to improve the performance of ensemble machine learning algorithms in a multicore environment, achieving a speed increase of 3.8x for testing datasets on quad-core CPUs.
- Developed and evaluated a Trojan localization algorithm to identify anomalous chip regions after detection, addressing a critical limitation of prior "detection-only" approaches.

Research Student, *Jeonbuk National University* **Jeonju, S.Korea | September 2021 – August 2022**

- Developed a model for predicting drug-target affinity using convolution-based self-attention mechanisms.
- Outperformed previous sequence-based or other approaches by 4% CI and demonstrated superior retention abilities.
- Handled large dataset of 117,657 drug-target interaction pairs, raw csv is 75 MB and preprocessed TFRecord file is 7.5 GB
- Utilized a multi-disciplinary approach that combined fields of science including chemistry, pharmacology, and biology.

Software Engineer, *YBC Services* (Remote) **Aldershot, UK | May 2021 – August 2021**

- Developed RESTful API based that served data to React and Flutter frontend used by human resource management software of YBC that handled over 1,000 concurrent users.
- Worked as a team leader with a task plan, blocker discussions, and project management, sped up development time by 13%

Software Engineer, *Bent Ray Technologies* **Lalitpur, Nepal | December 2018 – June 2021**

- Built and deployed 12+ production web applications after clients required rapid iteration across e-commerce and enterprise platforms.
- Designed and implemented REST APIs for a US-based staffing system, ensuring secure data exchange and maintainable backend architecture.

Microsoft Student Partner, *Microsoft Innovation Center Nepal* **Kathmandu, Nepal | July 2015 – June 2016**

- Represented Microsoft at the university level, organizing technical workshops and gaining early exposure to Azure cloud services and HoloLens development workflows.

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TECHNICAL SKILLS

Programming: Python, C/C++, Go, Bash
Machine / Deep Learning: scikit-learn, pandas, numpy, ML Ops, TensorFlow, Keras; CNNs, Transformers, Clustering
ML Security: Adversarial robustness toolbox (ART), Adversarial Learning, GAN, Gradient Masking, SHAP, LIME
Hardware Security: FPGA (Basys 3), power/EM side-channel data acquisition
Cybersecurity: Wireshark, tcpdump, Ghidra, Zeek, Binary Ninja
Tools & Platforms: Git, Docker, Singularity, SLURM, Jupyter
Web & APIs: RESTful services, microservices, Swagger, Postman
Systems & Networking: Linux, ns-3, distributed systems

PUBLICATIONS (Recent)

Ghimire, A., Zhang, J., Lingala, S. R., Alsulami, F., & Amsaad, F. **(2025). A Survey on Application of AI on Reverse Engineering for Software Analysis and Security**. IEEE Access.

Ghimire, A., Alkurdi, M., Rahman, M. T., Mohanty, S., & Amsaad, F. **(2025). A golden-free unsupervised ml-assisted security approach for detection of ic hardware trojans.** *ACM Journal on Emerging Technologies in Computing Systems*, 21(3), 1-22.

Ghimire, A., Ghajari, G., Gurung, K., Sah, L. K., & Amsaad, F. (**2025**). **Enhancing cybersecurity in critical infrastructure with llm-assisted explainable iot systems**. *1st International Conference on Secure IoT, Assured and Trusted Computing (SATC)*, pp. 1-5.

PROJECTS

Distributed Cloud–HPC Workflow Orchestration **2025**

- Built a **web-based orchestration system** after manual HPC job submission caused frequent configuration errors and lost results.
- Automated secure job execution across **Fry and Pitzer clusters** using SLURM, SSH/SCP, and database-backed job tracking, improving reliability of large-scale batch experiments.

TCP Performance Simulation using ns-3 **2024**

- Simulated TCP behavior under packet loss and delay conditions to diagnose throughput degradation observed in congested networks.
- Analyzed traffic traces to evaluate statistical and ML-based defenses against network-layer attacks.

Sentiment Analysis of Amazon Product Reviews using NLP and ML, **2023**

- A Built an NLP pipeline to classify customer sentiment after raw review data proved too noisy for manual analysis.
- Trained and evaluated ML models to categorize reviews into positive, negative, and neutral classes, supporting scalable customer insight extraction.

LEADERSHIP

Nepalese Student Association (Treasurer)
Wright State University **Aug 2023 - April 2024**

- Managed operational budget and logistics for a 200+ participant student organization.

RELEVANT COURSEWORK

• Machine Learning	• Deep Learning	• Hardware Security	• Distributed Computing
• Linux	• Big Data	• OOP	• Cloud Computing
• Data Structure and Algorithm	• Computer Network	• Embedded System	• Computer Architecture